MSC 201 **34202**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Ten independent trials each have success probability 0.3. X counts successes. What makes a binomial model appropriate?	A. Success probability exceeds 0.5. B. The expected count is an integer. C. Fixed trial count, two outcomes, independence, and constant success chance. D. Every trial must succeed. 10 pts
2. Independent tasks each succeed with probability 0.9. Which shortcut gives at least one failure among three tasks?	A. 1 minus the probability all three succeed. B. 1 minus the probability exactly one succeeds. C. 0.9 plus 0.9 plus 0.9. D. The probability all three fail. 10 pts
3. A population contains 4 observations with a sum of squared deviations from its mean of 36. What is the population variance?	A. 9. B. 36. C. 12. D. 3. 10 pts
4. Compare U(0,12) and U(20,32). What is true about their standard deviations?	A. The first is larger because it includes zero. B. They are equal. C. The second is larger because its mean is larger. D. Both standard deviations are zero. 10 pts
5. A box has 3 red and 7 blue tokens; 5 are drawn without replacement. Why is 4 red impossible?	A. Red probability must be zero. B. Only 3 red tokens exist. C. Half the population is drawn. D. The sample contains 5 tokens. 10 pts
6. A report gives sample SD=6 minutes and SE=1 minute. Which describes variation among individual observed times?	A. The confidence level. B. 7 minutes. C. 6 minutes. D. 1 minute. 10 pts
7. A fixed 5-kg container is added to a shipment of random item masses. What changes?	A. Expected total rises by 5 kg; total SD is unchanged. B. Both mean and SD rise by 5 kg. C. Total SD doubles. D. Expected total stays fixed. 10 pts

Test continues on the other side of the sheet.

MSC 201 **34202**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Students may use both laptops and tablets. Laptop use is 70%, tablet use 60%. Why is a two-slice pie chart inappropriate?	A. The categories overlap. B. A pie chart requires exactly three categories. C. Device use cannot be counted. D. Percentages can never exceed 50%. 10 pts
9. What equation describes a continuous model's 90th percentile k?	A. P(X<=k)=0.90. B. P(X=k)=0.90. C. The mean equals 0.90k. D. P(X>k)=0.90. 10 pts
10. One hundred valid 95% intervals are constructed in repeated sampling. Must exactly 95 cover the mean?	A. No; the coverage count itself can vary. B. Yes; any other count proves bias. C. No; all 100 must cover it. D. Yes; the confidence level fixes that count. 10 pts

34202

TEST PAGE 1

1.

C.

10 pts

2. A.

10 pts

3. A.

10 pts

4.

B.

10 pts

5.

B.

10 pts

6.

C.

10 pts

7. A.

10 pts

TEST PAGE 2

8. A.

10 pts

9. A.

10 pts

10. A.

10 pts

MSC 201 **93604**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Most files are small, with a few extremely large files. Which better describes a typical file's size?	A. The median. B. The maximum. C. The total size alone. D. The mean student ID. 10 pts
2. On a die, A is even and B is greater than 3. Which results satisfy both?	A. 5 only. B. 2 only. C. 2, 4, 5, 6. D. 4 and 6. 10 pts
3. A population is half faulty cables. A random sample of 6 is drawn without replacement. What faulty count is expected?	A. 6. B. 3. C. Exactly 3 in every sample. D. 0.5. 10 pts
4. Daily print totals are shown in a histogram instead of a time-series graph. What information is lost?	A. The existence of high totals. B. The number of daily observations. C. Which total occurred on which date. D. The approximate distribution of totals. 10 pts
5. An exponential density extends far right with diminishing height. What does that say about very long waits?	A. They must occur as often as short waits. B. Every long wait has probability 1. C. They are possible but less common than short waits. D. They are impossible once the mean is exceeded. 10 pts
6. A sample has 8 observations. Its sum of squared deviations is already supplied. Which divisor gives sample variance?	A. 64. B. The sample mean. C. 7. D. 8. 10 pts
7. Population SD doubles while sample size stays fixed. What happens to SE?	A. It quadruples. B. It stays unchanged. C. It doubles. D. It halves. 10 pts

Test continues on the other side of the sheet.

MSC 201 **93604**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. The course uses N(mean, SD). A technician expects readings from N(40,6) to cluster near 6. What is the best correction?	A. They cluster near 6; 40 is the sample size. B. The readings cluster near 40; 6 describes their spread. C. Every reading must equal 40. D. The model allows only the two values 6 and 40. 10 pts
9. A model describes the mass of one package. Is its SD automatically appropriate for the mean mass of 100 independent packages?	A. Yes; averages always vary more. B. No; the average has less spread. C. Yes; both are measured in grams. D. No; averages have no units. 10 pts
10. A symmetric interval is 8 to 12. What is the margin of error?	A. 4. B. 2. C. 10. D. 12. 10 pts

93604

TEST PAGE 1

1. A.

10 pts

2. D.

10 pts

3. B.

10 pts

4. C.

10 pts

5. C.

10 pts

6. C.

10 pts

7. C.

10 pts

TEST PAGE 2

8. B.

10 pts

9. B.

10 pts

10. B.

10 pts

MSC 201 **75802**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Population SD is known and the sample is large. What still needs checking for the usual mean interval?	A. Whether every observation equals the mean. B. Whether the mean is a whole number. C. Sampling quality and independence. D. Whether the interval includes zero. 10 pts
2. Mean is 30 and SD is 4. Which value has z=1.5?	A. 31.5. B. 26. C. 36. D. 45. 10 pts
3. Three reported percentages are 33.3%, 33.3%, and 33.3%. What most reasonably explains the 99.9% total?	A. Missing exactly one participant. B. Rounding. C. The data must be fabricated. D. Overlapping categories are certain. 10 pts
4. For comparison, sample SD and confidence are held fixed while n increases. What happens to a t mean interval's margin?	A. It stays unchanged. B. It increases because there are more observations. C. It becomes the population SD. D. It decreases as SE and the critical t value decrease. 10 pts
5. Data are 2, 4, 6, 8, 10. How do mean and median compare?	A. The median is the maximum. B. The mean is below the median. C. The mean exceeds the median. D. They are equal. 10 pts
6. A fixed fee increases by 2 dollars. What happens to net-gain mean and SD?	A. Mean is unchanged; SD rises. B. Both fall by 2. C. Both double. D. Mean falls by 2; SD is unchanged. 10 pts
7. P(A OR B)=0.75. What is the probability of neither?	A. 0.25. B. 0.50. C. 0.75. D. 1.75. 10 pts

Test continues on the other side of the sheet.

MSC 201 **75802**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A supplied probability table lists P(X=0),...,P(X=8). To find P(X<=2), which entries are needed?	A. Only the entry for 2. B. The expected value. C. The entries for 0, 1, and 2. D. The entries for 3 through 8. 10 pts
9. Individual masses have SD 20 g; means of 25 independent masses have SD 4 g. Which spread applies to one package?	A. 16 g. B. 20 g. C. 0.8 g. D. 4 g. 10 pts
10. A uniform interval centered at 10 doubles its width while keeping center 10. What happens?	A. Mean doubles; SD stays unchanged. B. Mean stays 10; SD doubles. C. Both double. D. Both stay unchanged. 10 pts

75802

TEST PAGE 1

1.

C.

10 pts

2.

C.

10 pts

3.

B.

10 pts

4.

D.

10 pts

5.

D.

10 pts

6.

D.

10 pts

7. A.

10 pts

TEST PAGE 2

8.

C.

10 pts

9.

B.

10 pts

10.

B.

10 pts

MSC 201 **56955**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Most waits are short, but a few extend far toward large values. What determines the skew direction?	A. The long tail toward large values. B. The number of categories. C. The side with the tallest bar alone. D. Whether the variable uses minutes. 10 pts
2. The ordered data are 2, 4, 4, 7, 20. What is the median?	A. 4. B. 7. C. 20. D. 6. 10 pts
3. A sum event and its equivalent mean event use correctly matched centers and SDs. How do their z-scores compare?	A. The sum z-score is n times larger. B. Their signs must be opposite. C. They are equal. D. The mean z-score is always zero. 10 pts
4. A symmetric confidence interval is 8 to 12. What point estimate is at its center?	A. 10. B. 12. C. 8. D. 4. 10 pts
5. Two distinct terminal paths describe exactly one head in two tosses. How is their combined probability found?	A. Add the two path probabilities. B. Ignore one because both describe one head. C. Subtract the smaller from the larger. D. Multiply the two path probabilities. 10 pts
6. Normal models share SD 4 but have means 10 and 14. Which has more probability below 8?	A. The mean-14 model. B. Both have probability 0.5. C. They have equal probabilities. D. The mean-10 model. 10 pts

Test continues on the other side of the sheet.

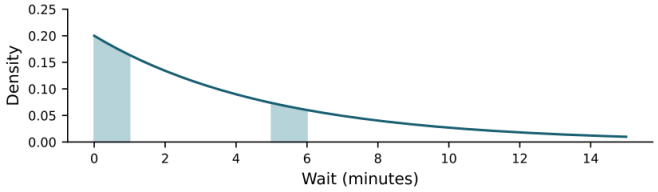
MSC 201 **56955**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. For an exponential wait, which one-minute interval has more probability? 	A. 5 to 6 minutes. B. Both have probability zero. C. 0 to 1 minute. D. They have equal probability. 10 pts
8. A survey records whether each student uses a service. Which summary directly answers how common use is?	A. The largest student ID. B. The mean student ID. C. The total of category labels chosen arbitrarily. D. The proportion of students reporting use. 10 pts
9. A question asks for the mass exceeded by only 10% of packages. What units should the answer have?	A. Probability units only. B. A percentage of sample size. C. No units because it is a z-score automatically. D. Mass units. 10 pts
10. A fixed fee increases by 2 dollars. What happens to net-gain mean and SD?	A. Mean falls by 2; SD is unchanged. B. Both double. C. Mean is unchanged; SD rises. D. Both fall by 2. 10 pts

56955

TEST PAGE 1

1. A.

10 pts

2. A.

10 pts

3. C.

10 pts

4. A.

10 pts

5. A.

10 pts

6. D.

10 pts

TEST PAGE 2

7. C.

10 pts

8. D.

10 pts

9. D.

10 pts

10. A.

10 pts

MSC 201 **85389**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Students may use both laptops and tablets. Laptop use is 70%, tablet use 60%. Why is a two-slice pie chart inappropriate?	A. Percentages can never exceed 50%. B. Device use cannot be counted. C. The categories overlap. D. A pie chart requires exactly three categories. 10 pts
2. A box has 8 red and 2 blue tokens; 5 are drawn without replacement. What is the minimum red count?	A. 2. B. 0. C. 5. D. 3. 10 pts
3. A question asks for the mass exceeded by only 10% of packages. What units should the answer have?	A. Probability units only. B. Mass units. C. No units because it is a z-score automatically. D. A percentage of sample size. 10 pts
4. The course uses N(mean, SD). A technician expects readings from N(40,6) to cluster near 6. What is the best correction?	A. Every reading must equal 40. B. They cluster near 6; 40 is the sample size. C. The model allows only the two values 6 and 40. D. The readings cluster near 40; 6 describes their spread. 10 pts
5. A sample has 8 observations. Its sum of squared deviations is already supplied. Which divisor gives sample variance?	A. 8. B. 7. C. 64. D. The sample mean. 10 pts
6. Sample SD is 6 minutes and n=9; sqrt(9)=3. What estimated SE is used?	A. 6 minutes. B. 18 minutes. C. 2 minutes. D. 3 minutes squared. 10 pts
7. To find the probability of A followed by B when the stages may be dependent, which expression fits?	A. P(A) times P(B given A). B. P(A) times P(B) without checking independence. C. P(A) minus P(B given A). D. P(A) plus P(B). 10 pts

Test continues on the other side of the sheet.

MSC 201 **85389**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Population mean is 20; SE of sample means is 2. An observed sample mean is 24. What standardized distance does it have?	A. 12. B. 4. C. 0.5. D. 2. 10 pts
9. Data are 2, 4, 6, 8, 10. How do mean and median compare?	A. The mean is below the median. B. The mean exceeds the median. C. They are equal. D. The median is the maximum. 10 pts
10. A wait is uniform from 0 to 12 minutes. What is $P(3<X<6)$?	A. 3/12. B. 9/12. C. 6/12. D. 3/6. 10 pts

85389

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3.

B.

10 pts

4.

D.

10 pts

5.

B.

10 pts

6.

C.

10 pts

7. A.

10 pts

TEST PAGE 2

8.

D.

10 pts

9.

C.

10 pts

10. A.

10 pts

MSC 201 **31664**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A normal model has mean 50 and SD 2. Which interval contains about 99.7%?	A. 44 to 56. B. 46 to 54. C. 50 to 56. D. 48 to 52. 10 pts
2. Durations are rounded to the nearest whole second. Ignoring tie conventions at boundaries, which exact interval reports as 4?	A. 4 to 5 seconds. B. 3.5 to 4.5 seconds. C. Only the exact point 4. D. 0 to 4 seconds. 10 pts
3. A sample of 6 has strong skewness and an extreme outlier. What is sensible about a t interval?	A. Six observations guarantee a normal sampling distribution. B. Its reliability deserves caution. C. The outlier must be removed without investigation. D. It is automatically reliable because t has heavy tails. 10 pts
4. What does a 95% confidence procedure mean in repeated appropriate sampling?	A. About 95% of intervals constructed this way contain the fixed population mean in the long run. B. 95% of observations lie in every interval. C. The sample mean equals the population mean 95% of the time. D. Every interval has 95% of its length below the mean. 10 pts
5. X counts failures in exactly 20 independent trials; Y counts arrivals during an hour with no fixed trial total. Which distinction matters?	A. Both must have the same maximum. B. X must be continuous. C. Y must be geometric. D. X fits a binomial setting; Y may fit a Poisson setting if its assumptions hold. 10 pts
6. A library surveys only people entering the library to estimate how often all students visit. What is the main concern?	A. Students who never visit are excluded. B. Randomly ordering the responses removes the concern. C. The sample must be too small. D. Every recorded visit count is wrong. 10 pts

Test continues on the other side of the sheet.

MSC 201 **31664**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. The ordered data are 1, 3, 5, 9. What is the median?	A. 9. B. 3. C. 4. D. 5. 10 pts
8. Ten independent trials each have success probability 0.3. X counts successes. What makes a binomial model appropriate?	A. Success probability exceeds 0.5. B. Every trial must succeed. C. Fixed trial count, two outcomes, independence, and constant success chance. D. The expected count is an integer. 10 pts
9. A model allows counts 0,1,2,... without a finite upper limit. Can it still be discrete?	A. No; a count must stop at 10. B. Yes; countably many separated values can be discrete. C. Only if every count has equal probability. D. No; all infinite possibilities are continuous. 10 pts
10. The course's binomial normal approximation check requires $np>5$ and $n(1-p)>5$. For $n=100$, $p=0.02$, which judgment fits?	A. It passes because n exceeds 30. B. It passes because p is positive. C. It fails because expected failures are 98. D. It fails because expected successes are only 2. 10 pts

31664

TEST PAGE 1

1. A.

10 pts

2. B.

10 pts

3. B.

10 pts

4. A.

10 pts

5. D.

10 pts

6. A.

10 pts

TEST PAGE 2

7. C.

10 pts

8. C.

10 pts

9. B.

10 pts

10. D.

10 pts

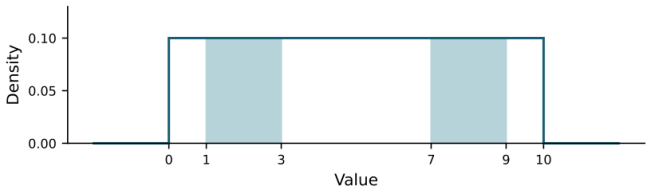
MSC 201 **53613**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Two successes in four trials can occur in several orders. Why must all those orders be considered?	A. The expected count is an order. B. The count event includes every order with two successes. C. Different orders are the same elementary outcome. D. Only the first successful order can occur. 10 pts
2. A recorded file size is far above the rest. What should happen before deleting it?	A. Replace it with zero automatically. B. Check whether it is an error or a genuine unusual file. C. Keep it only if the mean equals the median. D. Delete it because all outliers are mistakes. 10 pts
3. A proportion estimate is 0.40 with margin 0.05. What interval is this in percentages?	A. 5% to 40%. B. 40% to 50%. C. 35% to 45%. D. 39.95% to 40.05%. 10 pts
4. A grouped table reports 12 durations from 5 to less than 10 seconds. Can it tell exactly how many equal 7 seconds?	A. No; exact positions within the interval are hidden. B. Yes; none may equal 7. C. Yes; exactly half must equal 7. D. Yes; all 12 must equal 7. 10 pts
5. After five failures with constant independent p=0.2, what is the chance the next trial succeeds?	A. 0.2 divided by 5. B. 0.8. C. 0.2. D. 1. 10 pts
6. X is uniform from 0 to 10. Compare intervals 1 to 3 and 7 to 9. 	A. Their probabilities depend on the sample median. B. The second is more likely because its values are larger. C. The first is more likely because it is closer to zero. D. They have equal probability. 10 pts

Test continues on the other side of the sheet.

MSC 201 **53613**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A shipment contains 36 packages. Which variable answers the question of average mass per package in that shipment?	A. The total mass without dividing. B. The largest mass. C. The sample mean. D. A single randomly chosen mass. 10 pts
8. For a standard normal variable, compare P(Z<-1.4) and P(Z>1.4).	A. The right tail is larger. B. They are equal. C. Both equal 0.5. D. The left tail is larger. 10 pts
9. A fair six-sided die is rolled. What is the probability of a result greater than 4?	A. 1/6. B. 2/6. C. 4/6. D. 5/6. 10 pts
10. A 95% mean interval is 10 to 14 minutes. Does this say 95% of individual durations lie there?	A. Yes; the mean and individuals have identical spread. B. No; it instead guarantees every duration lies there. C. Yes; that is the meaning of any confidence interval. D. No; it estimates the population mean. 10 pts

53613

TEST PAGE 1

1.

B.

10 pts

2.

B.

10 pts

3.

C.

10 pts

4. A.

10 pts

5.

C.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

C.

10 pts

8.

B.

10 pts

9.

B.

10 pts

10.

D.

10 pts

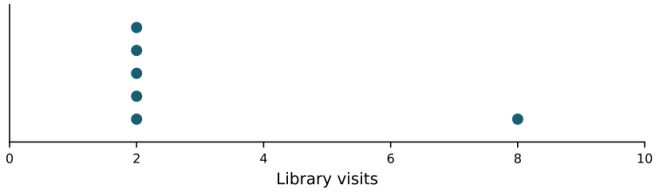
MSC 201 **89003**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Why is adding signed deviations from the sample mean unsuitable as a spread measure?	A. The mean is always the largest value. B. All deviations are positive. C. Positive and negative deviations cancel. D. It necessarily counts each value twice. 10 pts
2. A dot plot has five dots at 2 visits and one dot at 8 visits. What does the stack at 2 mean? 	A. One student reported 10 visits. B. The average is necessarily 2. C. Five students reported 8 visits each. D. Five students reported 2 visits each. 10 pts
3. Larger independent samples yield less variable means. Does this make individual packages less variable?	A. Only if more samples are plotted. B. No; population spread stays the same. C. Yes; individual SD becomes zero. D. Yes; every package moves toward the mean. 10 pts
4. The slides recommend plus-four when confidence is at least 90% and n>=10. Does n=24 at 85% confidence meet that recommendation?	A. Yes; n alone is sufficient. B. No; n must exceed 100. C. Yes; plus-four has no stated conditions. D. No; the confidence level is below the stated threshold. 10 pts
5. Probability is shifted from a low outcome to a higher outcome, with other probabilities unchanged. What happens to expected value?	A. It increases. B. It cannot change. C. It decreases. D. It must become the largest outcome. 10 pts
6. A uniform interval centered at 10 doubles its width while keeping center 10. What happens?	A. Both stay unchanged. B. Both double. C. Mean doubles; SD stays unchanged. D. Mean stays 10; SD doubles. 10 pts

Test continues on the other side of the sheet.

MSC 201 **89003**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A supplied probability table lists P(X=0),...,P(X=8). To find P(X<=2), which entries are needed?	A. The entries for 0, 1, and 2. B. The entries for 3 through 8. C. Only the entry for 2. D. The expected value. 10 pts
8. From the same sample, a 90% and 99% mean interval are built with the same method. Which statement fits?	A. Both widths must be identical. B. The 99% interval is wider but has the same center. C. The 90% interval is wider. D. The 99% interval has a different sample mean. 10 pts
9. Mean is 100, SD is 10, and the supplied 90th-percentile z is 1.28. What is the corresponding value?	A. 112.8. B. 101.28. C. 87.2. D. 12.8. 10 pts
10. P(A)=0.2 but P(A AND B)=0.3. Why is this impossible?	A. B must have probability 1. B. The overlap must always equal zero. C. The overlap cannot exceed the probability of A. D. A probability cannot be below 0.5. 10 pts

89003

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3.

B.

10 pts

4.

D.

10 pts

5. A.

10 pts

6.

D.

10 pts

TEST PAGE 2

7. A.

10 pts

8.

B.

10 pts

9. A.

10 pts

10.

C.

10 pts

MSC 201 **76220**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A bell-shaped distribution has mean 20 and SD 2. Approximately which interval contains 95%?	A. 0 to 20. B. 18 to 22. C. 16 to 24. D. 20 to 24. 10 pts
2. A proposed normal model gives substantial probability to negative durations. What is the concern?	A. Every duration must be negative. B. Normal distributions cannot model any positive variable. C. It assigns appreciable probability to impossible values. D. Its mean must be zero. 10 pts
3. A distribution gives P(X=0)=0.2 and P(X=1)=0.5, with only X=0,1,2 possible. What is P(X=2)?	A. 0.5. B. 0.7. C. 1.3. D. 0.3. 10 pts
4. A sum event and its equivalent mean event use correctly matched centers and SDs. How do their z-scores compare?	A. They are equal. B. The mean z-score is always zero. C. Their signs must be opposite. D. The sum z-score is n times larger. 10 pts
5. For a normal model, where is the 10th percentile?	A. At the largest possible value. B. Below the mean. C. At the mean. D. Above the mean. 10 pts
6. A supplied exponential model gives P(X>8)=0.20. What is P(X<=8)?	A. 1.20. B. 0.40. C. 0.20. D. 0.80. 10 pts

Test continues on the other side of the sheet.

MSC 201 **76220**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Two reasonable quartile methods give upper outlier fences of 18 and 20 for the same dataset. One observation is 35. Which conclusion agrees with both methods?	A. The observation is flagged as unusually high under both methods. B. Only the method with fence 18 flags the observation. C. The observation must be a recording error. D. The observation is not flagged under either method. 10 pts
8. For comparison, sample SD and confidence are held fixed while n increases. What happens to a t mean interval's margin?	A. It becomes the population SD. B. It decreases as SE and the critical t value decrease. C. It increases because there are more observations. D. It stays unchanged. 10 pts
9. X counts successes in 8 trials. Which proposed outcome is impossible?	A. X=0. B. X=9. C. X=8. D. X=4. 10 pts
10. A public dataset includes names beside individual survey answers. Which change best protects privacy while preserving group summaries?	A. Replace every answer with the overall mean. B. Remove identifying details before sharing. C. Publish only the names. D. Keep names and omit the study methods. 10 pts

76220

TEST PAGE 1

1.

C.

10 pts

2.

C.

10 pts

3.

D.

10 pts

4. A.

10 pts

5.

B.

10 pts

6.

D.

10 pts

TEST PAGE 2

7. A.

10 pts

8.

B.

10 pts

9.

B.

10 pts

10.

B.

10 pts

MSC 201 **31259**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A grader does not know which practice format each paper's author used. What problem does this reduce?	A. Students being absent from the population. B. All variation in student ability. C. The possibility of random assignment. D. Expectations influencing grading. 10 pts
2. Compare one U(0,12) wait with the average of 100 independent such waits. Which is less variable?	A. They have identical spread. B. The average must have zero spread. C. The individual wait. D. The average. 10 pts
3. From the same sample, a 90% and 99% mean interval are built with the same method. Which statement fits?	A. The 99% interval is wider but has the same center. B. The 90% interval is wider. C. Both widths must be identical. D. The 99% interval has a different sample mean. 10 pts
4. Mean is 100, SD is 10, and the supplied 90th-percentile z is 1.28. What is the corresponding value?	A. 12.8. B. 112.8. C. 87.2. D. 101.28. 10 pts
5. Outcomes 0 and 10 have probabilities 0.9 and 0.1. Why is their expected value not 5?	A. The larger value is impossible. B. The values cannot be averaged. C. The outcomes require unequal probability weights. D. Expected value always equals the most likely outcome. 10 pts
6. Which event is the complement of at least one failure in three trials?	A. At least two failures. B. No failures. C. Exactly one failure. D. All three failures. 10 pts
7. About 99.7% lies within 3 SD in a normal model. About what percent lies outside on both sides combined?	A. 99.7%. B. 3%. C. 0.3%. D. 0.15%. 10 pts

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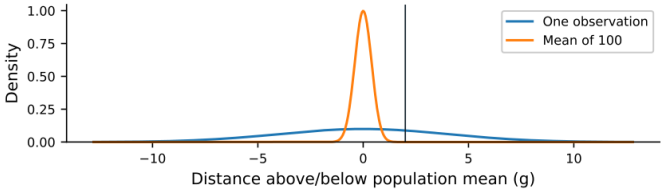
MSC 201 **31259**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. For a normal population, compare one mass exceeding mean+2 g with the mean of 100 independent masses exceeding the same cutoff. Which is more likely? 	A. Both have probability zero. B. The 100-mass mean event. C. They are equally likely. D. The individual mass event. 10 pts
9. For 2, 3, 3, 4, 18, how does the mean compare with the median?	A. They are equal. B. The mean is larger. C. No comparison is possible without a calculator. D. The mean is smaller. 10 pts
10. Durations are in seconds. Which units belong to variance and standard deviation, respectively?	A. Seconds squared; seconds. B. Percent; percent. C. Seconds; seconds squared. D. Seconds; no units. 10 pts

31259

TEST PAGE 1

1.

D.

10 pts

2.

D.

10 pts

3. A.

10 pts

4.

B.

10 pts

5.

C.

10 pts

6.

B.

10 pts

7.

C.

10 pts

TEST PAGE 2

8.

D.

10 pts

9.

B.

10 pts

10. A.

10 pts

MSC 201 **51741**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Among 200 recorded jobs, 30 were late. What does 30/200 describe?	A. A guarantee that the next job is late. B. The observed late-job relative frequency. C. The exact long-run probability under every possible condition. D. The number of late jobs that must occur tomorrow. 10 pts
2. Most waits are short, but a few extend far toward large values. What determines the skew direction?	A. The long tail toward large values. B. The side with the tallest bar alone. C. The number of categories. D. Whether the variable uses minutes. 10 pts
3. For a bell-shaped normal model, about 68% lies within 1 SD. About how much lies outside that central interval?	A. 95%. B. 68%. C. 16% in total. D. 32%. 10 pts
4. A report gives 30 successes among 120 observations. Which is the estimate of population success proportion?	A. 30. B. 0.25. C. 120. D. 90. 10 pts
5. A public dataset includes names beside individual survey answers. Which change best protects privacy while preserving group summaries?	A. Publish only the names. B. Replace every answer with the overall mean. C. Remove identifying details before sharing. D. Keep names and omit the study methods. 10 pts
6. Five print jobs contain 2, 3, 3, 5, and 7 pages. What is the mean number of pages per job?	A. 4. B. 20. C. 5. D. 3. 10 pts

Test continues on the other side of the sheet.

MSC 201 **51741**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Ten cables are sampled without replacement from 100,000. Why might binomial be a useful approximation?	A. No-replacement draws become exactly independent. B. All samples have the same faulty count. C. The faulty count is continuous. D. The small sample barely changes the population composition. 10 pts
8. A probability interval is widened to include its original interval plus a region of positive density. What happens to its probability?	A. It exceeds 1 automatically. B. It stays unchanged. C. It decreases. D. It increases. 10 pts
9. Population mean is 20; SE of sample means is 2. An observed sample mean is 24. What standardized distance does it have?	A. 4. B. 12. C. 2. D. 0.5. 10 pts
10. Uniform model A spans width 4; model B spans width 8. Which density is taller?	A. The one with the larger mean regardless of width. B. A. C. B. D. They have equal height. 10 pts

51741

TEST PAGE 1

1.

B.

10 pts

2. A.

10 pts

3.

D.

10 pts

4.

B.

10 pts

5.

C.

10 pts

6. A.

10 pts

TEST PAGE 2

7.

D.

10 pts

8.

D.

10 pts

9.

C.

10 pts

10.

B.

10 pts

MSC 201 **16334**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A population contains 4 observations with a sum of squared deviations from its mean of 36. What is the population variance?	A. 36. B. 3. C. 9. D. 12. 10 pts
2. Ten cables are sampled without replacement from 100,000. Why might binomial be a useful approximation?	A. The faulty count is continuous. B. All samples have the same faulty count. C. The small sample barely changes the population composition. D. No-replacement draws become exactly independent. 10 pts
3. Which Venn region is the complement of A OR B?	A. Only the overlap. B. Inside A only. C. Outside both circles. D. Every region inside either circle. 10 pts
4. Population SD is known and the sample is large. What still needs checking for the usual mean interval?	A. Whether the mean is a whole number. B. Whether every observation equals the mean. C. Sampling quality and independence. D. Whether the interval includes zero. 10 pts
5. Categories in a bar graph are reordered alphabetically. What changes?	A. The display order, but not category frequencies. B. Every relative frequency. C. The most frequent category. D. The number of observations in each category. 10 pts
6. For a bell-shaped normal model, about 68% lies within 1 SD. About how much lies outside that central interval?	A. 95%. B. 32%. C. 16% in total. D. 68%. 10 pts
7. An exponential density extends far right with diminishing height. What does that say about very long waits?	A. They must occur as often as short waits. B. They are impossible once the mean is exceeded. C. They are possible but less common than short waits. D. Every long wait has probability 1. 10 pts

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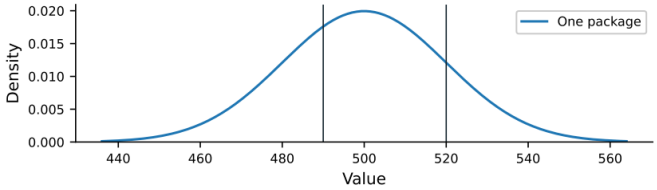
MSC 201 **16334**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A package is accepted if 490<mass<520. Which part of its normal curve should be shaded? 	A. Everything above 520. B. Both outside tails. C. Only the area between 490 and 520. D. Only the point 500. 10 pts
9. A mean is estimated from grouped intervals using midpoints. Why is it generally approximate?	A. Midpoints are never numerical values. B. Frequencies cannot be used in means. C. Actual observations need not equal the interval midpoints. D. The sample size is necessarily unknown. 10 pts
10. A sum distribution has mean 100 and SD 4. What total is 2 SD above its mean?	A. 102. B. 108. C. 200. D. 104. 10 pts

16334

TEST PAGE 1

1.

C.

10 pts

2.

C.

10 pts

3.

C.

10 pts

4.

C.

10 pts

5. A.

10 pts

6.

B.

10 pts

7.

C.

10 pts

TEST PAGE 2

8.

C.

10 pts

9.

C.

10 pts

10.

B.

10 pts

MSC 201 **28739**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A uniform wait ranges from 0 to 10. Only 10% exceed a cutoff. Where is that cutoff?	A. At 5. B. At 1. C. At 9. D. At 10. 10 pts
2. Game A pays 5 with certainty. Game B pays 0 or 10 with equal chance. What is true? 	A. They have equal expected payout; B has greater spread. B. A has greater spread. C. B has a larger expected payout. D. Both have zero spread. 10 pts
3. A supplied positive critical z leaves 0.025 to its right. What cumulative area lies to its left?	A. 0.025. B. 0.5. C. 0.975. D. 0.95. 10 pts
4. The plus-four estimate adds two successes and two failures. With 8 successes out of 40, what adjusted fraction is used?	A. 8/44. B. 10/40. C. 12/44. D. 10/44. 10 pts
5. Students may use both laptops and tablets. Laptop use is 70%, tablet use 60%. Why is a two-slice pie chart inappropriate?	A. A pie chart requires exactly three categories. B. Device use cannot be counted. C. Percentages can never exceed 50%. D. The categories overlap. 10 pts
6. A fixed 5-kg container is added to a shipment of random item masses. What changes?	A. Total SD doubles. B. Expected total rises by 5 kg; total SD is unchanged. C. Both mean and SD rise by 5 kg. D. Expected total stays fixed. 10 pts

Test continues on the other side of the sheet.

MSC 201 **28739**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. After a specific interval is computed, what is correct about its relation to a fixed population mean?	A. It either contains that mean or does not; 95% describes the procedure. B. The fixed mean changes to fit it with probability 95%. C. The mean occupies 95% of the interval. D. The endpoints must equal the smallest and largest observation. 10 pts
8. Which design permits an exact binomial count model with independent randomized draws?	A. Drawing most of a small population without replacement. B. Replacing and remixing each drawn token. C. Always stopping at first red. D. Drawing without replacement until the box is empty. 10 pts
9. Among 20 tablet users, 12 also use laptops. What is P(laptop given tablet) in this sample?	A. 8/20. B. 12 divided by all college students. C. 12/20. D. 20/12. 10 pts
10. A file is 60 KB; its group has mean 50 KB and SD 5 KB. How far above the mean is it?	A. 2 standard deviations. B. 10 standard deviations. C. 0.5 standard deviations. D. 12 standard deviations. 10 pts

28739

TEST PAGE 1

1.

C.

10 pts

2. A.

10 pts

3.

C.

10 pts

4.

D.

10 pts

5.

D.

10 pts

6.

B.

10 pts

TEST PAGE 2

7. A.

10 pts

8.

B.

10 pts

9.

C.

10 pts

10. A.

10 pts

MSC 201 **72667**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Q1 is 8 and Q3 is 14. What does the interval 8 to 14 describe approximately?	A. The lowest quarter only. B. The middle half of observations. C. Every observation. D. The mean's confidence interval. 10 pts
2. A payout variable is measured in dollars. What units belong to its variance?	A. Dollars squared. B. No units. C. Percent. D. Dollars. 10 pts
3. Independent tasks each succeed with probability 0.9. Which shortcut gives at least one failure among three tasks?	A. The probability all three fail. B. 1 minus the probability exactly one succeeds. C. 1 minus the probability all three succeed. D. 0.9 plus 0.9 plus 0.9. 10 pts
4. A frequency table lists 1 page for 8 jobs and 5 pages for 2 jobs. Why is (1+5)/2 inappropriate for mean pages per job?	A. It counts too many individual jobs. B. It ignores how often each value occurs. C. It gives extra weight to 1-page jobs. D. It is always the median. 10 pts
5. The course uses N(mean, SD). A technician expects readings from N(40,6) to cluster near 6. What is the best correction?	A. The model allows only the two values 6 and 40. B. The readings cluster near 40; 6 describes their spread. C. Every reading must equal 40. D. They cluster near 6; 40 is the sample size. 10 pts
6. What equation describes a continuous model's 90th percentile k?	A. The mean equals 0.90k. B. P(X<=k)=0.90. C. P(X=k)=0.90. D. P(X>k)=0.90. 10 pts
7. A uniform wait ranges from 0 to 10. Only 10% exceed a cutoff. Where is that cutoff?	A. At 9. B. At 1. C. At 5. D. At 10. 10 pts

Test continues on the other side of the sheet.

MSC 201 **72667**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Three reported percentages are 33.3%, 33.3%, and 33.3%. What most reasonably explains the 99.9% total?	A. Overlapping categories are certain. B. The data must be fabricated. C. Missing exactly one participant. D. Rounding. 10 pts
9. At fixed n and confidence, known population SD doubles. What happens to interval width?	A. It quadruples. B. It doubles. C. It halves. D. It stays unchanged. 10 pts
10. A population has mean 50. Where are unbiased random-sample means centered?	A. At the population SD. B. At 50 divided by sample size. C. At 50. D. At zero. 10 pts

72667

TEST PAGE 1

1.

B.

10 pts

2. A.

10 pts

3.

C.

10 pts

4.

B.

10 pts

5.

B.

10 pts

6.

B.

10 pts

7. A.

10 pts

TEST PAGE 2

8.

D.

10 pts

9.

B.

10 pts

10.

C.

10 pts

MSC 201 **96175**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A researcher takes 1,000 samples of 25 packages and plots each sample mean. What does one plotted value represent?	A. A population parameter that changes on each sample. B. The mass of one package. C. The mean of one 25-package sample. D. The mean of all 25,000 packages combined. 10 pts
2. A rate estimate rises from 40% to 45%. What is its increase in percentage points?	A. 0.05 percentage points. B. 12.5 percentage points. C. 45 percentage points. D. 5 percentage points. 10 pts
3. Mean is 30 and SD is 4. Which value has z=1.5?	A. 26. B. 31.5. C. 45. D. 36. 10 pts
4. For a normal model, where is the 10th percentile?	A. Below the mean. B. At the largest possible value. C. At the mean. D. Above the mean. 10 pts
5. A survey records whether each student uses a service. Which summary directly answers how common use is?	A. The mean student ID. B. The proportion of students reporting use. C. The largest student ID. D. The total of category labels chosen arbitrarily. 10 pts
6. Compare 4, 5, 6 with 1, 5, 9. Which statement is correct?	A. The first has more spread. B. They have equal means, but the second has more spread. C. The second has a larger mean. D. Equal means imply equal spread. 10 pts
7. A modified box plot marks 30 as a separate high outlier. Its upper whisker ends at 18. What does 18 represent?	A. The mean. B. The upper fence whether observed or not. C. The largest observed value not flagged as an outlier. D. The maximum of all observations. 10 pts

Test continues on the other side of the sheet.

MSC 201 **96175**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A machine's success chance steadily falls as it warms. Which binomial requirement is questionable?	A. A fixed number of trials. B. A numerical success count. C. Constant success probability across trials. D. There being two possible outcomes. 10 pts
9. F(3)=0.20 and F(7)=0.80 for a continuous variable. What is P(3<X<7)?	A. 0.20. B. 0.60. C. 1.00. D. 0.80. 10 pts
10. Two distinct terminal paths describe exactly one head in two tosses. How is their combined probability found?	A. Ignore one because both describe one head. B. Subtract the smaller from the larger. C. Multiply the two path probabilities. D. Add the two path probabilities. 10 pts

96175

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3.

D.

10 pts

4. A.

10 pts

5.

B.

10 pts

6.

B.

10 pts

7.

C.

10 pts

TEST PAGE 2

8.

C.

10 pts

9.

B.

10 pts

10.

D.

10 pts

MSC 201 **19468**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. All ten observations equal 5, so their median is 5. Are exactly half strictly below the median?	A. No; none are strictly below it. B. No; a median cannot exist with ties. C. Yes; exactly five are below 5. D. Yes; every median splits strict inequalities equally. 10 pts
2. A researcher deletes genuine results that disagree with the desired conclusion. What is the problem?	A. A smaller dataset guarantees less bias. B. Deleting inconvenient values always improves accuracy. C. Any unusual observation is automatically an error. D. The report no longer fairly represents the collected evidence. 10 pts
3. A proposed normal model gives substantial probability to negative durations. What is the concern?	A. Its mean must be zero. B. Normal distributions cannot model any positive variable. C. Every duration must be negative. D. It assigns appreciable probability to impossible values. 10 pts
4. A fair die is rolled. A means even; B means 6. Why are these events dependent?	A. A fair die makes all events independent. B. Both are separate die rolls. C. Knowing B guarantees A, changing its probability. D. They cannot happen together. 10 pts
5. X counts broken cables in independent checks. Can broken be called success in a binomial model?	A. Only if no cables are broken. B. Only if more than half are broken. C. Yes; success means the counted outcome. D. No; success must be desirable. 10 pts
6. A distribution has equal peaks near 2 and 8 and is symmetric around 5. What follows?	A. Its mode must be uniquely 5. B. Symmetry does not require a single peak. C. A mean cannot exist. D. It must be right-skewed. 10 pts

Test continues on the other side of the sheet.

MSC 201 **19468**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. One hundred valid 95% intervals are constructed in repeated sampling. Must exactly 95 cover the mean?	A. No; all 100 must cover it. B. No; the coverage count itself can vary. C. Yes; any other count proves bias. D. Yes; the confidence level fixes that count. 10 pts
8. The 95th percentile of a sample-mean distribution is 14. What does it describe?	A. Every sample mean is 14. B. The unknown mean has become 14. C. 95% of all individuals must be at most 14. D. About 95% of repeated sample means are at most 14. 10 pts
9. The model changes from U(0,10) to U(0,20). What happens to P(X<5)?	A. It halves. B. It doubles. C. It becomes 1. D. It stays unchanged. 10 pts
10. A normal model's mean rises by 3 with SD fixed. What happens to its 90th percentile?	A. It falls by 3. B. It stays unchanged. C. It rises by 90%. D. It rises by 3. 10 pts

19468

TEST PAGE 1

1. A.

10 pts

2. D.

10 pts

3. D.

10 pts

4. C.

10 pts

5. C.

10 pts

6. B.

10 pts

TEST PAGE 2

7. B.

10 pts

8. D.

10 pts

9. A.

10 pts

10. D.

10 pts

MSC 201 **26535**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A mean is estimated from grouped intervals using midpoints. Why is it generally approximate?	A. Actual observations need not equal the interval midpoints. B. Midpoints are never numerical values. C. The sample size is necessarily unknown. D. Frequencies cannot be used in means. 10 pts
2. About 99.7% lies within 3 SD in a normal model. About what percent lies outside on both sides combined?	A. 0.15%. B. 0.3%. C. 99.7%. D. 3%. 10 pts
3. A first branch has probability 0.4; its next conditional branch is 0.5. What is the path probability?	A. 0.2. B. 0.9. C. 0.8. D. 0.1. 10 pts
4. The course's ordinary normal-approximation interval check requires more than 5 observed successes and more than 5 failures. Which sample passes?	A. 5 successes and 95 failures. B. 2 successes and 98 failures. C. 8 successes and 12 failures. D. 20 successes and 0 failures. 10 pts
5. What equation describes a continuous model's 90th percentile k?	A. P(X<=k)=0.90. B. P(X=k)=0.90. C. The mean equals 0.90k. D. P(X>k)=0.90. 10 pts
6. Four mutually exclusive, exhaustive categories have reported proportions 0.20, 0.30, 0.40, and 0.25. What needs checking?	A. Proportions must be whole numbers. B. Their total exceeds 1. C. Each category must have proportion 0.25. D. No category may exceed 0.20. 10 pts
7. A population has mean 50. Where are unbiased random-sample means centered?	A. At 50. B. At the population SD. C. At zero. D. At 50 divided by sample size. 10 pts

Test continues on the other side of the sheet.

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A sample variance is 9 minutes squared. What is its standard deviation?	A. 3 minutes squared. B. 81 minutes. C. 9 minutes. D. 3 minutes. 10 pts
9. A fixed fee increases by 2 dollars. What happens to net-gain mean and SD?	A. Both fall by 2. B. Mean is unchanged; SD rises. C. Both double. D. Mean falls by 2; SD is unchanged. 10 pts
10. Arrival rates sharply rise at lunchtime. What is questionable about a single exponential waiting model over the whole day?	A. A constant rate across the whole period. B. Nonnegative waiting times. C. Recording waits in minutes. D. Allowing a long wait. 10 pts

26535

TEST PAGE 1

1. A.

10 pts

2. B.

10 pts

3. A.

10 pts

4. C.

10 pts

5. A.

10 pts

6. B.

10 pts

7. A.

10 pts

TEST PAGE 2

8. D.

10 pts

9. D.

10 pts

10. A.

10 pts

MSC 201 **41008**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A proposed density dips below zero over an interval. What is wrong?	A. Every valid density must be symmetric. B. It would assign negative area to that interval. C. A density must be flat everywhere. D. Its maximum must equal 1. 10 pts
2. Population SD doubles while sample size stays fixed. What happens to SE?	A. It halves. B. It stays unchanged. C. It quadruples. D. It doubles. 10 pts
3. A box has 4 red and 6 blue tokens; 3 are drawn without replacement. What red counts are possible?	A. 1, 2, 3, 4. B. Only 0 or 3. C. 0, 1, 2, 3. D. 0 through 6. 10 pts
4. Everyone tries format A first and format B second. What could explain better second scores besides format B?	A. The lack of any treatment order. B. A guarantee of equal prior knowledge. C. Practice from the first attempt. D. Random sampling of all students. 10 pts
5. A model predicts 5% of items fall outside specifications. Among 200 items, how many are expected outside?	A. 5. B. 40. C. 10. D. Exactly 10 must occur. 10 pts
6. Durations are in seconds. Which units belong to variance and standard deviation, respectively?	A. Percent; percent. B. Seconds; no units. C. Seconds squared; seconds. D. Seconds; seconds squared. 10 pts
7. A model allows counts 0,1,2,... without a finite upper limit. Can it still be discrete?	A. Only if every count has equal probability. B. Yes; countably many separated values can be discrete. C. No; all infinite possibilities are continuous. D. No; a count must stop at 10. 10 pts

Test continues on the other side of the sheet.

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A strongly skewed variable is converted to z-scores. Does it become normal?	A. No; standardization shifts and rescales without removing skew. B. Yes; subtracting the mean removes the tail. C. Yes; every z-score distribution is standard normal. D. No; z-scores cannot be negative. 10 pts
9. A score is reported at approximately the 90th percentile. What kind of information does that give?	A. That 90% of exam answers were correct. B. Its position relative to other scores. C. A score of exactly 90 points. D. The class mean being 90. 10 pts
10. A sample has 2 successes and 98 failures. Does a large total alone justify the ordinary normal-approximation proportion interval?	A. No; proportions cannot be estimated at all. B. Yes; failures are numerous. C. No; the observed success count is too sparse. D. Yes; n=100 guarantees every condition. 10 pts

41008

TEST PAGE 1

1.

B.

10 pts

2.

D.

10 pts

3.

C.

10 pts

4.

C.

10 pts

5.

C.

10 pts

6.

C.

10 pts

7.

B.

10 pts

TEST PAGE 2

8. A.

10 pts

9.

B.

10 pts

10.

C.

10 pts

MSC 201 **37318**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. The number of independent items rises from 25 to 100. What happens to expected sum and SD of sum?	A. Expected sum quadruples; SD doubles. B. Expected sum stays fixed; SD halves. C. Both quadruple. D. Both halve. 10 pts
2. A proposed event probability is 1.15. What is wrong?	A. A probability cannot exceed 1. B. All probabilities must be integers. C. Only complements may exceed 1. D. Probabilities must always equal 0.5. 10 pts
3. A fixed fee increases by 2 dollars. What happens to net-gain mean and SD?	A. Mean is unchanged; SD rises. B. Both fall by 2. C. Mean falls by 2; SD is unchanged. D. Both double. 10 pts
4. Which question targets a population proportion?	A. What is the SD of file sizes? B. What fraction of jobs finish on time? C. What is the median number of pages? D. What is the mean duration of jobs? 10 pts
5. Durations are in seconds. Which units belong to variance and standard deviation, respectively?	A. Percent; percent. B. Seconds; seconds squared. C. Seconds squared; seconds. D. Seconds; no units. 10 pts
6. A recorded file size is far above the rest. What should happen before deleting it?	A. Keep it only if the mean equals the median. B. Delete it because all outliers are mistakes. C. Replace it with zero automatically. D. Check whether it is an error or a genuine unusual file. 10 pts
7. A continuous model has F(9)=0.80. What percent exceeds 9?	A. 80%. B. 9%. C. 100%. D. 20%. 10 pts

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MSC 201 **37318**

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8. Mean is 30 and SD is 4. Which value has z=1.5?	A. 45. B. 36. C. 26. D. 31.5. 10 pts
9. A room at 20 degrees Celsius is compared with one at 10 degrees Celsius. Which statement is justified?	A. The first has twice the absolute temperature. B. The first is 10 degrees warmer on that scale. C. Celsius readings cannot be subtracted. D. The first is twice as comfortable for every person. 10 pts
10. P(Z>1.2)=0.1151. By symmetry, what is P(Z<-1.2)?	A. -0.1151. B. 0.5. C. 0.8849. D. 0.1151. 10 pts

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TEST PAGE 1

1. A.

10 pts

2. A.

10 pts

3. C.

10 pts

4. B.

10 pts

5. C.

10 pts

6. D.

10 pts

7. D.

10 pts

TEST PAGE 2

8. B.

10 pts

9. B.

10 pts

10. D.

10 pts

MSC 201 **12509**

Diagnostic Exam

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Requests follow a constant-rate Poisson process at 4 per hour. What is the mean waiting time to the next request?	A. 15 minutes. B. 4 minutes. C. 60 minutes. D. 240 minutes. 10 pts
2. A stemplot key says stem 2, leaf 5 means 25 KB. Stem 3 has leaves 0, 2, 2. Which values does it show?	A. 300, 320, 320 KB. B. 30, 32 KB only. C. 3, 0, 2, 2 KB. D. 30, 32, 32 KB. 10 pts
3. Data are 2, 4, 6, 8, 10. How do mean and median compare?	A. They are equal. B. The median is the maximum. C. The mean exceeds the median. D. The mean is below the median. 10 pts
4. A symmetric interval is 8 to 12. What is the margin of error?	A. 2. B. 4. C. 12. D. 10. 10 pts
5. P(X>50)=0.50 and P(X>60)=0.10. Given X>50, what is P(X>60)?	A. 0.60. B. 0.20. C. 0.10. D. 0.40. 10 pts
6. P(A OR B)=0.75. What is the probability of neither?	A. 0.50. B. 0.75. C. 0.25. D. 1.75. 10 pts
7. Compare 4, 5, 6 with 1, 5, 9. Which statement is correct?	A. The first has more spread. B. Equal means imply equal spread. C. They have equal means, but the second has more spread. D. The second has a larger mean. 10 pts

Test continues on the other side of the sheet.

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8. An exponential population has mean 5. Which is generally more unusual under independent sampling: one duration above 7 or 100 durations averaging above 7?	A. The single duration above 7. B. The average of 100 above 7. C. They have equal probability. D. Neither can occur. 10 pts
9. A supplied normal table gives $P(Z<1.2)=0.8849$. What is $P(Z>1.2)$?	A. 0.44245. B. 0.8849. C. 0.1151. D. 1.8849. 10 pts
10. Which variable is a count rather than a waiting time?	A. Requests received in the next five minutes. B. Exact time between two requests. C. Duration of a request. D. Minutes until the next request. 10 pts

12509

TEST PAGE 1

1. A.

10 pts

2. D.

10 pts

3. A.

10 pts

4. A.

10 pts

5. B.

10 pts

6. C.

10 pts

7. C.

10 pts

TEST PAGE 2

8. B.

10 pts

9. C.

10 pts

10. A.

10 pts